

# Tutorial 10: PCA and Factor Analysis

ENVX2001 – Applied Statistical Methods

Semester 1

## PCA and Factor Analysis

Students in class gave “Characteristics of a good ecology lecturer”. They scored from 1-10 on a range of characteristics:

- accessible
- attractive
- Wide range of topics
- Do teaching research
- Dress well
- Complex issues
- Example exam questions
- Notes before
- humour
- Easy marking
- Move around lecture
- Detailed lecture notes
- Clear objective to lecture
- Solicit questions
- Use textbook closely
- Focus on primary research
- Use their own research
- Teach in team
- Understand principles of teaching
- Use pointers
- respected

The students also indicated whether they were:

- a gender (male/female)
- studied mainly a system (marine/terrestrial)
- studied mainly an organismal group (plants/animals)

We are going to use principal components analysis and factor analysis to interpret this dataset.

## Setup

Load these packages to make the plots look better:

```
CODE
library(EFAtools)
library(ggplot2)
library(ggfortify)
```

## Load the data

First open your dataset:

```
CODE
lecturers <- read.csv("data/lecturers.csv", header = TRUE)
```

## Correlation matrix

Let's first do a correlation matrix:

```
CODE
cor(lecturers[, 8:28])
```

```
OUTPUT
```

|                         | accessible  | attractive  | wide_range   |
|-------------------------|-------------|-------------|--------------|
| accessible              | 1.00000000  | -0.61876084 | 0.190062066  |
| attractive              | -0.61876084 | 1.00000000  | -0.011570169 |
| wide_range              | 0.19006207  | -0.01157017 | 1.000000000  |
| teaching_based_research | 0.24021638  | -0.21382943 | 0.462870058  |
| Dress_nicely            | -0.53198037 | 0.81887024  | -0.022848836 |
| focus_on_complex_issues | 0.28252341  | -0.04447540 | 0.567443460  |
| gives_example           | -0.23669223 | 0.23435837  | 0.065815857  |
| gives_out_note          | -0.31173854 | 0.36819405  | -0.187661524 |
| sense_of_humour         | -0.29068273 | 0.33747896  | 0.081503385  |
| mark_easily             | -0.34976811 | 0.44361281  | -0.158545138 |
| moves_around            | -0.23371225 | 0.18475768  | -0.087292839 |
| offers_lecture_notes    | -0.25762452 | 0.28256570  | -0.070931605 |
| clear_objective         | -0.21819811 | 0.14978520  | -0.176518448 |
| solicit_questions       | 0.10349600  | -0.07667948 | -0.022123679 |
| text_book               | 0.03170769  | 0.02849978  | -0.241108100 |
| primary_research        | 0.14147306  | -0.06238516 | -0.009628388 |
| own_research            | -0.01089281 | 0.15175696  | 0.157013029  |
| teach_in_team           | 0.09004589  | -0.13904104 | 0.005163136  |
| understand_principles   | -0.08521319 | -0.01080764 | -0.097173242 |
| use_a_pointer           | -0.51717009 | 0.25526667  | -0.131935123 |
| well_respected          | -0.12789662 | 0.04222022  | 0.037698097  |
| teaching_based_research |             |             |              |
| accessible              |             | 0.24021638  | -0.531980374 |
| attractive              |             | -0.21382943 | 0.818870242  |
| wide_range              |             | 0.46287006  | -0.022848836 |
| teaching_based_research |             | 1.00000000  | -0.079662581 |
| Dress_nicely            |             | -0.07966258 | 1.000000000  |
| focus_on_complex_issues |             | 0.36612054  | -0.074522662 |

|                                                        |              |              |              |
|--------------------------------------------------------|--------------|--------------|--------------|
| gives_example                                          | -0.23626944  | 0.207509283  |              |
| gives_out_note                                         | -0.13154529  | 0.349298764  |              |
| sense_of_humour                                        | -0.37115541  | 0.326981372  |              |
| mark_easily                                            | -0.26802282  | 0.293513890  |              |
| moves_around                                           | 0.35915941   | 0.175612992  |              |
| offers_lecture_notes                                   | -0.26062206  | 0.304730151  |              |
| clear_objective                                        | -0.10137698  | 0.162176313  |              |
| solicit_questions                                      | 0.06997014   | -0.047353486 |              |
| text_book                                              | -0.02511112  | 0.065661794  |              |
| primary_research                                       | 0.08627931   | -0.125765210 |              |
| own_research                                           | 0.24450347   | 0.131114596  |              |
| teach_in_team                                          | 0.05621459   | -0.081891992 |              |
| understand_principles                                  | 0.13360929   | 0.003830791  |              |
| use_a_pointer                                          | -0.40503932  | 0.246189390  |              |
| well_respected                                         | -0.21574299  | 0.082434616  |              |
| focus_on_complex_issues gives_example gives_out_note   |              |              |              |
| accessible                                             | 0.28252341   | -0.23669223  | -0.31173854  |
| attractive                                             | -0.04447540  | 0.23435837   | 0.36819405   |
| wide_range                                             | 0.56744346   | 0.06581586   | -0.18766152  |
| teaching_based_research                                | 0.36612054   | -0.23626944  | -0.13154529  |
| Dress_nicely                                           | -0.07452266  | 0.20750928   | 0.34929876   |
| focus_on_complex_issues                                | 1.00000000   | -0.15606162  | -0.31138949  |
| gives_example                                          | -0.15606162  | 1.00000000   | 0.41459959   |
| gives_out_note                                         | -0.31138949  | 0.41459959   | 1.00000000   |
| sense_of_humour                                        | -0.07005663  | 0.22759892   | 0.25028755   |
| mark_easily                                            | -0.18475242  | 0.18880641   | 0.39259669   |
| moves_around                                           | 0.06116819   | -0.15496603  | 0.26746010   |
| offers_lecture_notes                                   | 0.04438636   | 0.67139954   | 0.32650197   |
| clear_objective                                        | -0.05740760  | 0.22152350   | 0.49329825   |
| solicit_questions                                      | 0.12547982   | -0.10034117  | 0.37306647   |
| text_book                                              | -0.14904151  | 0.23189356   | 0.41705241   |
| primary_research                                       | 0.33824172   | -0.05502391  | -0.00258233  |
| own_research                                           | 0.21007669   | -0.03858742  | 0.11486585   |
| teach_in_team                                          | -0.05016868  | 0.22073735   | 0.07101795   |
| understand_principles                                  | 0.12582565   | 0.06367067   | 0.09564742   |
| use_a_pointer                                          | -0.19722690  | 0.29421612   | 0.42798936   |
| well_respected                                         | -0.15321258  | 0.20022401   | 0.24197769   |
| sense_of_humour mark_easily moves_around               |              |              |              |
| accessible                                             | -0.29068273  | -0.349768108 | -0.233712249 |
| attractive                                             | 0.33747896   | 0.443612809  | 0.184757685  |
| wide_range                                             | 0.08150339   | -0.158545138 | -0.087292839 |
| teaching_based_research                                | -0.37115541  | -0.268022817 | 0.359159406  |
| Dress_nicely                                           | 0.32698137   | 0.293513890  | 0.175612992  |
| focus_on_complex_issues                                | -0.07005663  | -0.184752419 | 0.061168190  |
| gives_example                                          | 0.22759892   | 0.188806413  | -0.154966026 |
| gives_out_note                                         | 0.25028755   | 0.392596691  | 0.267460102  |
| sense_of_humour                                        | 1.00000000   | 0.279195853  | 0.147981262  |
| mark_easily                                            | 0.27919585   | 1.000000000  | -0.102624026 |
| moves_around                                           | 0.14798126   | -0.102624026 | 1.000000000  |
| offers_lecture_notes                                   | 0.40350521   | 0.004540766  | 0.116586008  |
| clear_objective                                        | 0.30494267   | 0.241714172  | 0.328029643  |
| solicit_questions                                      | 0.22196924   | 0.221507736  | 0.357039548  |
| text_book                                              | 0.11515826   | 0.493885950  | 0.018944976  |
| primary_research                                       | -0.11402253  | 0.013988221  | 0.201361759  |
| own_research                                           | 0.19400676   | -0.136905883 | 0.521788320  |
| teach_in_team                                          | -0.13889755  | -0.029214646 | -0.002012914 |
| understand_principles                                  | 0.08973153   | 0.029511600  | 0.281044416  |
| use_a_pointer                                          | 0.60147995   | 0.200483903  | 0.247735287  |
| well_respected                                         | 0.31777004   | 0.204840208  | 0.107275533  |
| offers_lecture_notes clear_objective solicit_questions |              |              |              |
| accessible                                             | -0.257624516 | -0.21819811  | 0.103495996  |
| attractive                                             | 0.282565698  | 0.14978520   | -0.076679483 |
| wide_range                                             | -0.070931605 | -0.17651845  | -0.022123679 |
| teaching_based_research                                | -0.260622062 | -0.10137698  | 0.069970142  |
| Dress_nicely                                           | 0.304730151  | 0.16217631   | -0.047353486 |

|                                                   |              |              |              |
|---------------------------------------------------|--------------|--------------|--------------|
| focus_on_complex_issues                           | 0.044386363  | -0.05740760  | 0.125479819  |
| gives_example                                     | 0.671399539  | 0.22152350   | -0.100341166 |
| gives_out_note                                    | 0.326501972  | 0.49329825   | 0.373066471  |
| sense_of_humour                                   | 0.403505211  | 0.30494267   | 0.221969236  |
| mark_easily                                       | 0.004540766  | 0.24171417   | 0.221507736  |
| moves_around                                      | 0.116586008  | 0.32802964   | 0.357039548  |
| offers_lecture_notes                              | 1.000000000  | 0.51135418   | 0.134906607  |
| clear_objective                                   | 0.511354179  | 1.000000000  | 0.667857876  |
| solicit_questions                                 | 0.134906607  | 0.66785788   | 1.000000000  |
| text_book                                         | 0.241763186  | 0.50985104   | 0.514856144  |
| primary_research                                  | 0.027238866  | -0.05933422  | 0.008953955  |
| own_research                                      | 0.014724611  | 0.04037952   | 0.313649567  |
| teach_in_team                                     | 0.134589361  | 0.09431599   | 0.129502715  |
| understand_principles                             | 0.204933977  | 0.52773521   | 0.525180942  |
| use_a_pointer                                     | 0.475656687  | 0.48740938   | 0.245720968  |
| well_respected                                    | 0.173541917  | 0.20806935   | 0.217669413  |
| text_book primary_research own_research           |              |              |              |
| accessible                                        | 0.031707686  | 0.141473062  | -0.01089281  |
| attractive                                        | 0.028499784  | -0.062385161 | 0.15175696   |
| wide_range                                        | -0.241108100 | -0.009628388 | 0.15701303   |
| teaching_based_research                           | -0.025111119 | 0.086279306  | 0.24450347   |
| Dress_nicely                                      | 0.065661794  | -0.125765210 | 0.13111460   |
| focus_on_complex_issues                           | -0.149041509 | 0.338241715  | 0.21007669   |
| gives_example                                     | 0.231893556  | -0.055023907 | -0.03858742  |
| gives_out_note                                    | 0.417052411  | -0.002582330 | 0.11486585   |
| sense_of_humour                                   | 0.115158256  | -0.114022533 | 0.19400676   |
| mark_easily                                       | 0.493885950  | 0.013988221  | -0.13690588  |
| moves_around                                      | 0.018944976  | 0.201361759  | 0.52178832   |
| offers_lecture_notes                              | 0.241763186  | 0.027238866  | 0.01472461   |
| clear_objective                                   | 0.509851042  | -0.059334220 | 0.04037952   |
| solicit_questions                                 | 0.514856144  | 0.008953955  | 0.31364957   |
| text_book                                         | 1.000000000  | -0.092231864 | -0.11384268  |
| primary_research                                  | -0.092231864 | 1.000000000  | 0.35299282   |
| own_research                                      | -0.113842676 | 0.352992824  | 1.000000000  |
| teach_in_team                                     | 0.176402014  | -0.085933891 | -0.08668898  |
| understand_principles                             | 0.287773155  | 0.017218953  | 0.06370238   |
| use_a_pointer                                     | 0.001337384  | -0.149851010 | 0.15399945   |
| well_respected                                    | 0.274222528  | 0.016210854  | -0.10140227  |
| teach_in_team understand_principles use_a_pointer |              |              |              |
| accessible                                        | 0.090045886  | -0.085213191 | -0.517170092 |
| attractive                                        | -0.139041035 | -0.010807636 | 0.255266673  |
| wide_range                                        | 0.005163136  | -0.097173242 | -0.131935123 |
| teaching_based_research                           | 0.056214590  | 0.133609295  | -0.405039321 |
| Dress_nicely                                      | -0.081891992 | 0.003830791  | 0.246189390  |
| focus_on_complex_issues                           | -0.050168682 | 0.125825652  | -0.197226905 |
| gives_example                                     | 0.220737347  | 0.063670673  | 0.294216117  |
| gives_out_note                                    | 0.071017950  | 0.095647419  | 0.427989355  |
| sense_of_humour                                   | -0.138897546 | 0.089731525  | 0.601479950  |
| mark_easily                                       | -0.029214646 | 0.029511600  | 0.200483903  |
| moves_around                                      | -0.002012914 | 0.281044416  | 0.247735287  |
| offers_lecture_notes                              | 0.134589361  | 0.204933977  | 0.475656687  |
| clear_objective                                   | 0.094315990  | 0.527735214  | 0.487409381  |
| solicit_questions                                 | 0.129502715  | 0.525180942  | 0.245720968  |
| text_book                                         | 0.176402014  | 0.287773155  | 0.001337384  |
| primary_research                                  | -0.085933891 | 0.017218953  | -0.149851010 |
| own_research                                      | -0.086688981 | 0.063702380  | 0.153999453  |
| teach_in_team                                     | 1.000000000  | 0.101857142  | -0.016777680 |
| understand_principles                             | 0.101857142  | 1.000000000  | 0.104631048  |
| use_a_pointer                                     | -0.016777680 | 0.104631048  | 1.000000000  |
| well_respected                                    | 0.545915684  | 0.207782811  | 0.318541502  |
| well_respected                                    |              |              |              |
| accessible                                        | -0.12789662  |              |              |
| attractive                                        | 0.04222022   |              |              |
| wide_range                                        | 0.03769810   |              |              |
| teaching_based_research                           | -0.21574299  |              |              |

|                         |             |
|-------------------------|-------------|
| Dress_nicely            | 0.08243462  |
| focus_on_complex_issues | -0.15321258 |
| gives_example           | 0.20022401  |
| gives_out_note          | 0.24197769  |
| sense_of_humour         | 0.31777004  |
| mark_easily             | 0.20484021  |
| moves_around            | 0.10727553  |
| offers_lecture_notes    | 0.17354192  |
| clear_objective         | 0.20806935  |
| solicit_questions       | 0.21766941  |
| text_book               | 0.27422253  |
| primary_research        | 0.01621085  |
| own_research            | -0.10140227 |
| teach_in_team           | 0.54591568  |
| understand_principles   | 0.20778281  |
| use_a_pointer           | 0.31854150  |
| well_respected          | 1.00000000  |

CODE

```
Corrmatrix <- cor(lecturers[, 8:28])
```

## Bartlett's Test of Sphericity

Now we can do the Bartlett's Test of Sphericity. This test compares the correlation matrix to an identity matrix. If it is significant, it is worth doing a PCA.

CODE

```
BARTLETT(Corrmatrix, N = 34, cor_method = c("pearson"))
```

OUTPUT

✓ The Bartlett's test of sphericity was significant at an alpha level of .05.  
These data are probably suitable for factor analysis.

$\chi^2(210) = 355.07, p < .001$

## Principal Components Analysis

Note: to make this a PCA based on a correlation matrix, we have to scale the variables, hence `scale = TRUE`. There are two main principal components functions, but they are very similar. Note `prcomp` calls the loadings "rotations", not to be confused with rotations below.

CODE

```
pca1 <- prcomp(lecturers[, 8:28], scale = TRUE)
pca2 <- princomp(lecturers[, 8:28], cor = TRUE)
```

CODE

```
summary(pca1)
```

#### OUTPUT

Importance of components:

|                        | PC1     | PC2     | PC3     | PC4     | PC5     | PC6     | PC7     |
|------------------------|---------|---------|---------|---------|---------|---------|---------|
| Standard deviation     | 2.2206  | 1.6616  | 1.5324  | 1.29807 | 1.19561 | 1.12098 | 1.08311 |
| Proportion of Variance | 0.2348  | 0.1315  | 0.1118  | 0.08024 | 0.06807 | 0.05984 | 0.05586 |
| Cumulative Proportion  | 0.2348  | 0.3663  | 0.4781  | 0.55833 | 0.62641 | 0.68624 | 0.74211 |
|                        | PC8     | PC9     | PC10    | PC11    | PC12    | PC13    | PC14    |
| Standard deviation     | 1.0737  | 0.95681 | 0.79545 | 0.73119 | 0.7055  | 0.65580 | 0.50887 |
| Proportion of Variance | 0.0549  | 0.04359 | 0.03013 | 0.02546 | 0.0237  | 0.02048 | 0.01233 |
| Cumulative Proportion  | 0.7970  | 0.84060 | 0.87073 | 0.89619 | 0.9199  | 0.94037 | 0.95270 |
|                        | PC15    | PC16    | PC17    | PC18    | PC19    | PC20    | PC21    |
| Standard deviation     | 0.49817 | 0.47309 | 0.42640 | 0.36970 | 0.32493 | 0.24020 | 0.19872 |
| Proportion of Variance | 0.01182 | 0.01066 | 0.00866 | 0.00651 | 0.00503 | 0.00275 | 0.00188 |
| Cumulative Proportion  | 0.96452 | 0.97518 | 0.98384 | 0.99034 | 0.99537 | 0.99812 | 1.00000 |

#### CODE

```
summary(pca2)
```

#### OUTPUT

Importance of components:

|                        | Comp.1      | Comp.2      | Comp.3      | Comp.4      | Comp.5     |
|------------------------|-------------|-------------|-------------|-------------|------------|
| Standard deviation     | 2.2205654   | 1.6615887   | 1.5324030   | 1.29806969  | 1.19560985 |
| Proportion of Variance | 0.2348053   | 0.1314703   | 0.1118219   | 0.08023738  | 0.06807062 |
| Cumulative Proportion  | 0.2348053   | 0.3662756   | 0.4780975   | 0.55833484  | 0.62640546 |
|                        | Comp.6      | Comp.7      | Comp.8      | Comp.9      | Comp.10    |
| Standard deviation     | 1.12097809  | 1.08311410  | 1.07371597  | 0.95680927  | 0.79544930 |
| Proportion of Variance | 0.05983771  | 0.05586363  | 0.05489838  | 0.04359448  | 0.03013046 |
| Cumulative Proportion  | 0.68624317  | 0.74210679  | 0.79700517  | 0.84059965  | 0.87073010 |
|                        | Comp.11     | Comp.12     | Comp.13     | Comp.14     | Comp.15    |
| Standard deviation     | 0.73119377  | 0.70551609  | 0.65579857  | 0.50886890  | 0.49817027 |
| Proportion of Variance | 0.02545925  | 0.02370252  | 0.02047961  | 0.01233084  | 0.01181779 |
| Cumulative Proportion  | 0.89618936  | 0.91989188  | 0.94037149  | 0.95270232  | 0.96452011 |
|                        | Comp.16     | Comp.17     | Comp.18     | Comp.19     |            |
| Standard deviation     | 0.4730918   | 0.426395520 | 0.369702631 | 0.324933440 |            |
| Proportion of Variance | 0.0106579   | 0.008657769 | 0.006508573 | 0.005027702 |            |
| Cumulative Proportion  | 0.9751780   | 0.983835781 | 0.990344355 | 0.995372057 |            |
|                        | Comp.20     | Comp.21     |             |             |            |
| Standard deviation     | 0.240198751 | 0.198724365 |             |             |            |
| Proportion of Variance | 0.002747402 | 0.001880542 |             |             |            |
| Cumulative Proportion  | 0.998119458 | 1.000000000 |             |             |            |

## Loadings

Let's look at the loadings. Called "rotations" in `prcomp` and "loadings" in `princomp`. They are the Pearson's correlation between that variable and that Principal Component.

#### CODE

```
pca1
```

#### OUTPUT

Standard deviations (1, .., p=21):

```
[1] 2.2205654 1.6615887 1.5324030 1.2980697 1.1956099 1.1209781 1.0831141
[8] 1.0737160 0.9568093 0.7954493 0.7311938 0.7055161 0.6557986 0.5088689
[15] 0.4981703 0.4730918 0.4263955 0.3697026 0.3249334 0.2401988 0.1987244
```

Rotation (n x k) = (21 x 21):

|                         | PC1          | PC2         | PC3          | PC4          |
|-------------------------|--------------|-------------|--------------|--------------|
| accessible              | -0.27105784  | 0.21481141  | 0.22284903   | -0.109380126 |
| attractive              | 0.25953819   | -0.19001404 | -0.35322829  | 0.047967149  |
| wide_range              | -0.11796047  | 0.14523503  | -0.23536689  | -0.465316942 |
| teaching.based_research | -0.17807186  | 0.33523672  | -0.16973235  | -0.005540905 |
| Dress_nicely            | 0.24737662   | -0.15550598 | -0.32565089  | 0.019172145  |
| focus_on_complex_issues | -0.13467331  | 0.28837561  | -0.24412817  | -0.297540766 |
| gives_example           | 0.22797978   | -0.11739164 | 0.06984178   | -0.437201342 |
| gives_out_note          | 0.32056620   | 0.04438472  | 0.02749136   | 0.100499687  |
| sense_of_humour         | 0.27564472   | -0.03120757 | -0.13957405  | -0.145704311 |
| mark_easily             | 0.23686285   | -0.11243202 | 0.08113616   | 0.196601185  |
| moves_around            | 0.13236700   | 0.32827473  | -0.27087444  | 0.213497862  |
| offers_lecture_notes    | 0.28307512   | 0.02357872  | -0.01627849  | -0.372830743 |
| clear_objective         | 0.31573398   | 0.26378859  | 0.12826236   | 0.066194937  |
| solicit_questions       | 0.18991164   | 0.43659484  | 0.13380240   | 0.160366349  |
| text_book               | 0.21299001   | 0.16712512  | 0.33485831   | 0.124899445  |
| primary_research        | -0.05608411  | 0.19206706  | -0.16387602  | 0.001745603  |
| own_research            | 0.04318371   | 0.28387549  | -0.37836892  | 0.056442158  |
| teach_in_team           | 0.04818137   | 0.11021896  | 0.29523823   | -0.327405642 |
| understand_principles   | 0.14556568   | 0.33936721  | 0.10777339   | 0.053017840  |
| use_a_pointer           | 0.32019196   | -0.03811643 | -0.08101559  | -0.080227102 |
| well_respected          | 0.19384039   | 0.06977679  | 0.22029921   | -0.263146206 |
|                         | PC5          | PC6         | PC7          | PC8          |
| accessible              | -0.006862732 | 0.20175545  | 0.075192964  | 0.13471983   |
| attractive              | -0.304217305 | -0.05278985 | -0.006014137 | -0.01346850  |
| wide_range              | -0.251311171 | 0.08634713  | -0.331523146 | 0.07096692   |
| teaching.based_research | -0.307853355 | -0.22290900 | -0.040540304 | -0.27328747  |
| Dress_nicely            | -0.299180706 | -0.12573075 | -0.046516486 | -0.12380132  |
| focus_on_complex_issues | -0.170654964 | 0.29340602  | -0.050872762 | 0.07474147   |
| gives_example           | -0.075647764 | 0.06855598  | 0.364935010  | -0.17131172  |
| gives_out_note          | -0.125013318 | -0.08366732 | 0.223406707  | 0.02145305   |
| sense_of_humour         | 0.216056466  | 0.21464296  | -0.328598912 | 0.27397871   |
| mark_easily             | -0.421810061 | 0.15983061  | -0.046406386 | 0.38018642   |
| moves_around            | 0.166526391  | -0.35024271 | 0.055092321  | -0.07707096  |
| offers_lecture_notes    | 0.146235397  | 0.18757528  | 0.287000528  | -0.27400475  |
| clear_objective         | 0.060342195  | 0.17777324  | -0.032682046 | -0.21837583  |
| solicit_questions       | -0.005900941 | 0.14588578  | -0.182965113 | 0.09074671   |
| text_book               | -0.356204508 | 0.15361788  | 0.084096939  | 0.02338965   |
| primary_research        | 0.039054189  | 0.06897414  | 0.592731543  | 0.45913721   |
| own_research            | 0.156844077  | -0.09496398 | 0.160942737  | 0.20826388   |
| teach_in_team           | -0.096207102 | -0.54758870 | 0.037625674  | 0.05473642   |
| understand_principles   | 0.010243773  | 0.05372028  | -0.114475017 | -0.27603888  |
| use_a_pointer           | 0.411769112  | 0.01616452  | -0.180288938 | 0.05917537   |
| well_respected          | 0.013159486  | -0.40377696 | -0.183055505 | 0.39655405   |
|                         | PC9          | PC10        | PC11         | PC12         |
| accessible              | 0.23711499   | 0.41979852  | 0.03945146   | 0.148881566  |
| attractive              | -0.16438427  | 0.20365082  | 0.06904042   | -0.107432225 |
| wide_range              | 0.22155409   | -0.26438749 | -0.03259079  | -0.018048262 |
| teaching.based_research | 0.19889412   | -0.24510749 | -0.11135378  | 0.247899071  |
| Dress_nicely            | -0.10406494  | 0.38639604  | 0.08377430   | 0.098387712  |
| focus_on_complex_issues | -0.28979913  | 0.00432150  | 0.36226348   | -0.066562002 |
| gives_example           | 0.19510676   | -0.15342441 | -0.35973985  | -0.159175284 |
| gives_out_note          | 0.41895610   | -0.21289394 | 0.20038838   | 0.006019332  |
| sense_of_humour         | 0.09730489   | 0.19787809  | -0.29410062  | 0.309619454  |
| mark_easily             | -0.04288340  | -0.25885827 | -0.05713892  | -0.143735223 |
| moves_around            | 0.02114182   | -0.04788447 | 0.02682634   | 0.353171999  |
| offers_lecture_notes    | -0.07931457  | 0.16874959  | 0.11255136   | 0.207814735  |
| clear_objective         | -0.04989256  | -0.08180409 | 0.29595319   | 0.009127372  |
| solicit_questions       | 0.10754501   | 0.13028684  | 0.14132328   | -0.276862219 |
| text_book               | 0.12628767   | 0.17521009  | -0.12180066  | 0.267854799  |
| primary_research        | -0.30348754  | -0.18921946 | 0.01954047   | 0.138689169  |
| own_research            | 0.28543090   | 0.27973695  | -0.29907250  | -0.427500246 |
| teach_in_team           | -0.07446630  | 0.20427502  | 0.21413563   | -0.322896702 |
| understand_principles   | -0.49690454  | -0.08895650 | -0.50033599  | -0.189676003 |
| use_a_pointer           | 0.09757553   | -0.27912944 | 0.22992547   | -0.115886008 |

|                         |              |              |              |              |
|-------------------------|--------------|--------------|--------------|--------------|
| well_respected          | -0.19000184  | -0.03498965  | -0.08926599  | 0.276488059  |
|                         | PC13         | PC14         | PC15         | PC16         |
| accessible              | 0.29231904   | 0.29580819   | -0.222218786 | 0.095682695  |
| attractive              | 0.10512874   | 0.20004636   | 0.069413634  | -0.326657310 |
| wide_range              | 0.14339377   | -0.15578216  | -0.042633744 | -0.346118318 |
| teaching_based_research | -0.16129758  | -0.08569727  | -0.291561886 | 0.304862272  |
| Dress_nicely            | 0.22024728   | -0.29025496  | -0.196064822 | 0.331319175  |
| focus_on_complex_issues | -0.11651678  | 0.21121417   | 0.380088085  | 0.275036237  |
| gives_example           | -0.02596566  | 0.13003779   | 0.073844050  | -0.048615414 |
| gives_out_note          | 0.52366782   | 0.22815294   | 0.150396690  | 0.187752138  |
| sense_of_humour         | -0.09102492  | 0.24787583   | -0.269328590 | 0.077361879  |
| mark_easily             | -0.30618081  | 0.26528469   | -0.239118638 | 0.067806707  |
| moves_around            | -0.19413727  | 0.41519824   | 0.163481854  | -0.233951595 |
| offers_lecture_notes    | -0.18626080  | -0.01943147  | -0.008128194 | -0.028997162 |
| clear_objective         | 0.02299500   | -0.21161788  | -0.392005665 | -0.377275931 |
| solicit_questions       | 0.07959209   | -0.06601352  | 0.067535709  | -0.098815936 |
| text_book               | -0.31596633  | -0.28217199  | 0.346092878  | 0.063696048  |
| primary_research        | 0.11269387   | -0.18656539  | -0.263215477 | -0.012502540 |
| own_research            | -0.14671997  | -0.23213427  | 0.082887276  | 0.001508984  |
| teach_in_team           | -0.23420220  | 0.19002827   | -0.271548974 | 0.056217287  |
| understand_principles   | 0.29771815   | 0.14826763   | -0.020146024 | 0.175848894  |
| use_a_pointer           | -0.10543967  | -0.11603812  | -0.008762183 | 0.433779829  |
| well_respected          | 0.22357927   | -0.22236358  | 0.251201419  | -0.060389007 |
|                         | PC17         | PC18         | PC19         | PC20         |
| accessible              | -0.487523780 | -0.092943639 | 0.11623660   | -0.08184741  |
| attractive              | -0.106937105 | -0.195239242 | 0.45086924   | -0.09762660  |
| wide_range              | 0.009616636  | 0.068156323  | 0.05963088   | -0.34244915  |
| teaching_based_research | 0.006376281  | 0.018683190  | 0.22919635   | 0.23688191   |
| Dress_nicely            | -0.124236446 | 0.174946567  | -0.38290347  | -0.11111538  |
| focus_on_complex_issues | 0.057689702  | -0.197551507 | -0.21480326  | 0.19658872   |
| gives_example           | -0.206970055 | 0.030539850  | -0.40647428  | -0.01600380  |
| gives_out_note          | 0.321857496  | -0.107588372 | 0.08121002   | 0.10250939   |
| sense_of_humour         | 0.430141107  | -0.113398698 | -0.08853507  | 0.02641527   |
| mark_easily             | -0.269651092 | 0.167631153  | 0.01976224   | 0.19779113   |
| moves_around            | -0.207902119 | 0.101506396  | -0.24054414  | -0.20539739  |
| offers_lecture_notes    | 0.024256519  | 0.371837654  | 0.41986631   | 0.20173823   |
| clear_objective         | -0.097333800 | -0.419634434 | -0.21848656  | 0.23314508   |
| solicit_questions       | 0.094290088  | 0.606166863  | -0.05758082  | -0.04550053  |
| text_book               | 0.072364215  | -0.254638772 | 0.07071928   | -0.35855399  |
| primary_research        | 0.130404141  | 0.041767487  | 0.01625913   | -0.25697656  |
| own_research            | -0.068753059 | -0.196939862 | 0.08026083   | 0.24801010   |
| teach_in_team           | 0.247146896  | -0.095095343 | 0.00645715   | -0.19776650  |
| understand_principles   | -0.001605969 | -0.075193184 | 0.12987739   | -0.13337875  |
| use_a_pointer           | -0.352414119 | -0.113600108 | 0.19774694   | -0.34608373  |
| well_respected          | -0.231089402 | 0.008802903  | 0.03736378   | 0.36974689   |
|                         | PC21         |              |              |              |
| accessible              | -0.002932735 |              |              |              |
| attractive              | 0.405148279  |              |              |              |
| wide_range              | -0.308435374 |              |              |              |
| teaching_based_research | 0.335322467  |              |              |              |
| Dress_nicely            | -0.150318850 |              |              |              |
| focus_on_complex_issues | 0.045604055  |              |              |              |
| gives_example           | 0.338040434  |              |              |              |
| gives_out_note          | -0.184254445 |              |              |              |
| sense_of_humour         | 0.144536843  |              |              |              |
| mark_easily             | -0.277239849 |              |              |              |
| moves_around            | -0.151918843 |              |              |              |
| offers_lecture_notes    | -0.289119982 |              |              |              |
| clear_objective         | -0.008051690 |              |              |              |
| solicit_questions       | 0.367111373  |              |              |              |
| text_book               | -0.026709294 |              |              |              |
| primary_research        | 0.123144467  |              |              |              |
| own_research            | -0.206478451 |              |              |              |
| teach_in_team           | -0.055864220 |              |              |              |
| understand_principles   | -0.183080778 |              |              |              |



|                |             |
|----------------|-------------|
| use_a_pointer  | 0.121125177 |
| well_respected | 0.079347988 |

CODE

loadings(pca2)

OUTPUT

Loadings:

|                         | Comp.1  | Comp.2  | Comp.3  | Comp.4  | Comp.5  | Comp.6  | Comp.7  | Comp.8 |
|-------------------------|---------|---------|---------|---------|---------|---------|---------|--------|
| accessible              | 0.271   | 0.215   | 0.223   | 0.109   |         | 0.202   |         | 0.135  |
| attractive              | -0.260  | -0.190  | -0.353  |         | 0.304   |         |         |        |
| wide_range              | 0.118   | 0.145   | -0.235  | 0.465   | 0.251   |         | -0.332  |        |
| teaching_based_research | 0.178   | 0.335   | -0.170  |         | 0.308   | -0.223  |         | -0.273 |
| Dress_nicely            | -0.247  | -0.156  | -0.326  |         | 0.299   | -0.126  |         | -0.124 |
| focus_on_complex_issues | 0.135   | 0.288   | -0.244  | 0.298   | 0.171   | 0.293   |         |        |
| gives_example           | -0.228  | -0.117  |         | 0.437   |         |         | 0.365   | -0.171 |
| gives_out_note          | -0.321  |         |         | -0.100  | 0.125   |         | 0.223   |        |
| sense_of_humour         | -0.276  |         | -0.140  | 0.146   | -0.216  | 0.215   | -0.329  | 0.274  |
| mark_easily             | -0.237  | -0.112  |         | -0.197  | 0.422   | 0.160   |         | 0.380  |
| moves_around            | -0.132  | 0.328   | -0.271  | -0.213  | -0.167  | -0.350  |         |        |
| offers_lecture_notes    | -0.283  |         |         | 0.373   | -0.146  | 0.188   | 0.287   | -0.274 |
| clear_objective         | -0.316  | 0.264   | 0.128   |         |         | 0.178   |         | -0.218 |
| solicit_questions       | -0.190  | 0.437   | 0.134   | -0.160  |         | 0.146   | -0.183  |        |
| text_book               | -0.213  | 0.167   | 0.335   | -0.125  | 0.356   | 0.154   |         |        |
| primary_research        |         | 0.192   | -0.164  |         |         |         | 0.593   | 0.459  |
| own_research            |         | 0.284   | -0.378  |         | -0.157  |         | 0.161   | 0.208  |
| teach_in_team           |         | 0.110   | 0.295   | 0.327   |         | -0.548  |         |        |
| understand_principles   | -0.146  | 0.339   | 0.108   |         |         |         | -0.114  | -0.276 |
| use_a_pointer           | -0.320  |         |         |         | -0.412  |         | -0.180  |        |
| well_respected          | -0.194  |         | 0.220   | 0.263   |         | -0.404  | -0.183  | 0.397  |
|                         | Comp.9  | Comp.10 | Comp.11 | Comp.12 | Comp.13 | Comp.14 | Comp.15 |        |
| accessible              | 0.237   | 0.420   |         | 0.149   | 0.292   | 0.296   | 0.222   |        |
| attractive              | -0.164  | 0.204   |         | -0.107  | 0.105   | 0.200   |         |        |
| wide_range              | 0.222   | -0.264  |         |         | 0.143   | -0.156  |         |        |
| teaching_based_research | 0.199   | -0.245  | -0.111  | 0.248   | -0.161  |         | 0.292   |        |
| Dress_nicely            | -0.104  | 0.386   |         |         | 0.220   | -0.290  | 0.196   |        |
| focus_on_complex_issues | -0.290  |         | 0.362   |         | -0.117  | 0.211   | -0.380  |        |
| gives_example           | 0.195   | -0.153  | -0.360  | -0.159  |         | 0.130   |         |        |
| gives_out_note          | 0.419   | -0.213  | 0.200   |         | 0.524   | 0.228   | -0.150  |        |
| sense_of_humour         |         | 0.198   | -0.294  | 0.310   |         | 0.248   | 0.269   |        |
| mark_easily             |         | -0.259  |         | -0.144  | -0.306  | 0.265   | 0.239   |        |
| moves_around            |         |         |         | 0.353   | -0.194  | 0.415   | -0.163  |        |
| offers_lecture_notes    |         | 0.169   | 0.113   | 0.208   | -0.186  |         |         |        |
| clear_objective         |         |         | 0.296   |         |         | -0.212  | 0.392   |        |
| solicit_questions       | 0.108   | 0.130   | 0.141   | -0.277  |         |         |         |        |
| text_book               | 0.126   | 0.175   | -0.122  | 0.268   | -0.316  | -0.282  | -0.346  |        |
| primary_research        | -0.303  | -0.189  |         | 0.139   | 0.113   | -0.187  | 0.263   |        |
| own_research            | 0.285   | 0.280   | -0.299  | -0.428  | -0.147  | -0.232  |         |        |
| teach_in_team           |         | 0.204   | 0.214   | -0.323  | -0.234  | 0.190   | 0.272   |        |
| understand_principles   | -0.497  |         | -0.500  | -0.190  | 0.298   | 0.148   |         |        |
| use_a_pointer           |         | -0.279  | 0.230   | -0.116  | -0.105  | -0.116  |         |        |
| well_respected          | -0.190  |         |         | 0.276   | 0.224   | -0.222  | -0.251  |        |
|                         | Comp.16 | Comp.17 | Comp.18 | Comp.19 | Comp.20 | Comp.21 |         |        |
| accessible              |         | 0.488   |         | 0.116   |         |         |         |        |
| attractive              | -0.327  | 0.107   | 0.195   | 0.451   |         | -0.405  |         |        |
| wide_range              | -0.346  |         |         |         | 0.342   | 0.308   |         |        |
| teaching_based_research | 0.305   |         |         | 0.229   | -0.237  | -0.335  |         |        |
| Dress_nicely            | 0.331   | 0.124   | -0.175  | -0.383  | 0.111   | 0.150   |         |        |
| focus_on_complex_issues | 0.275   |         | 0.198   | -0.215  | -0.197  |         |         |        |
| gives_example           |         | 0.207   |         | -0.406  |         | -0.338  |         |        |
| gives_out_note          | 0.188   | -0.322  | 0.108   |         | -0.103  | 0.184   |         |        |

|                       |        |        |        |        |        |        |        |        |  |
|-----------------------|--------|--------|--------|--------|--------|--------|--------|--------|--|
| sense_of_humour       |        | -0.430 | 0.113  |        |        |        |        | -0.145 |  |
| mark_easily           |        | 0.270  | -0.168 |        |        |        | -0.198 | 0.277  |  |
| moves_around          | -0.234 | 0.208  | -0.102 | -0.241 | 0.205  | 0.152  |        |        |  |
| offers_lecture_notes  |        |        | -0.372 | 0.420  | -0.202 | 0.289  |        |        |  |
| clear_objective       | -0.377 |        | 0.420  | -0.218 | -0.233 |        |        |        |  |
| solicit_questions     |        |        | -0.606 |        |        |        |        | -0.367 |  |
| text_book             |        |        | 0.255  |        |        | 0.359  |        |        |  |
| primary_research      |        | -0.130 |        |        |        | 0.257  | -0.123 |        |  |
| own_research          |        |        | 0.197  |        |        | -0.248 | 0.206  |        |  |
| teach_in_team         |        | -0.247 |        |        |        | 0.198  |        |        |  |
| understand_principles | 0.176  |        |        |        | 0.130  | 0.133  | 0.183  |        |  |
| use_a_pointer         | 0.434  | 0.352  | 0.114  | 0.198  | 0.346  | -0.121 |        |        |  |
| well_respected        |        | 0.231  |        |        |        | -0.370 |        |        |  |

|                |        |        |        |        |        |        |        |        |        |
|----------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
|                | Comp.1 | Comp.2 | Comp.3 | Comp.4 | Comp.5 | Comp.6 | Comp.7 | Comp.8 | Comp.9 |
| SS loadings    | 1.000  | 1.000  | 1.000  | 1.000  | 1.000  | 1.000  | 1.000  | 1.000  | 1.000  |
| Proportion Var | 0.048  | 0.048  | 0.048  | 0.048  | 0.048  | 0.048  | 0.048  | 0.048  | 0.048  |
| Cumulative Var | 0.048  | 0.095  | 0.143  | 0.190  | 0.238  | 0.286  | 0.333  | 0.381  | 0.429  |

|                |         |         |         |         |         |         |         |         |
|----------------|---------|---------|---------|---------|---------|---------|---------|---------|
|                | Comp.10 | Comp.11 | Comp.12 | Comp.13 | Comp.14 | Comp.15 | Comp.16 | Comp.17 |
| SS loadings    | 1.000   | 1.000   | 1.000   | 1.000   | 1.000   | 1.000   | 1.000   | 1.000   |
| Proportion Var | 0.048   | 0.048   | 0.048   | 0.048   | 0.048   | 0.048   | 0.048   | 0.048   |
| Cumulative Var | 0.476   | 0.524   | 0.571   | 0.619   | 0.667   | 0.714   | 0.762   | 0.810   |

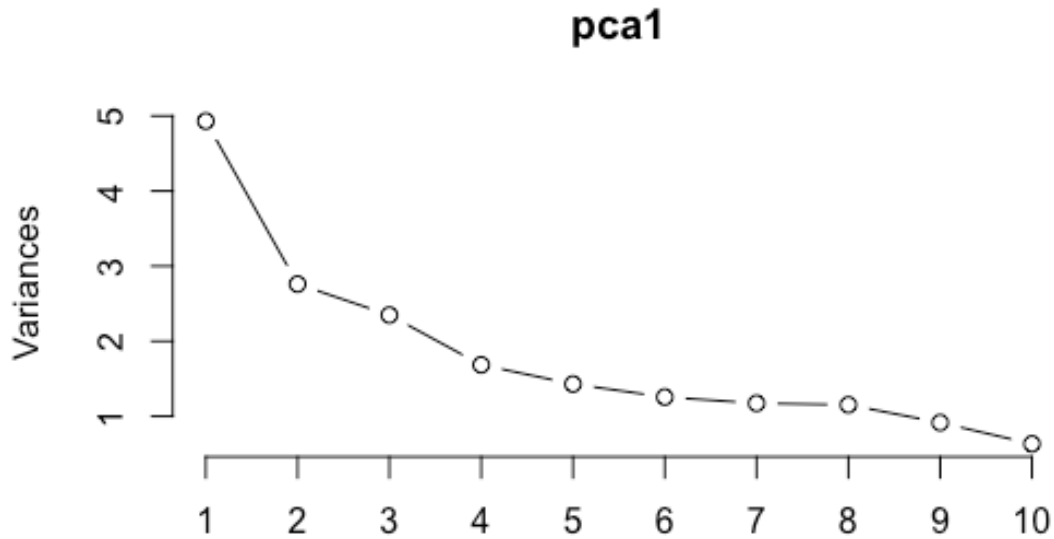
|                |         |         |         |         |
|----------------|---------|---------|---------|---------|
|                | Comp.18 | Comp.19 | Comp.20 | Comp.21 |
| SS loadings    | 1.000   | 1.000   | 1.000   | 1.000   |
| Proportion Var | 0.048   | 0.048   | 0.048   | 0.048   |
| Cumulative Var | 0.857   | 0.905   | 0.952   | 1.000   |

## Screeplot

To do a screeplot, follow the commands below. Note this is the standard deviations, which are just square root of the variances or eigenvalues.

CODE

```
screeplot(pca1, type = "lines")
```



## Principal Component Scores

To get your principal components scores for plotting and analysis, do the following:

CODE

```
pca1$x
```

OUTPUT

|       | PC1          | PC2         | PC3        | PC4        | PC5         | PC6          |
|-------|--------------|-------------|------------|------------|-------------|--------------|
| [1,]  | 0.834364400  | -0.44633674 | -0.7389322 | 0.6505980  | 0.67162071  | -0.059426515 |
| [2,]  | -0.057591233 | 2.43259604  | 0.3643921  | 0.1044273  | -1.35364156 | -1.513379283 |
| [3,]  | -1.414356172 | -0.85486631 | 0.2591048  | -0.7950895 | 0.39445964  | -0.362282532 |
| [4,]  | -3.007651543 | -1.54426710 | 1.8013306  | -1.4495208 | -0.06900951 | -0.378437639 |
| [5,]  | -0.913357854 | 0.15572053  | 1.9852660  | 0.2647465  | 0.05455313  | -0.190589811 |
| [6,]  | 2.715375811  | -0.61458292 | 1.2581973  | 0.6291478  | 1.66664928  | 0.980101358  |
| [7,]  | -3.260274779 | -1.46022078 | -1.3182136 | -1.5445289 | 2.59801832  | 1.373084699  |
| [8,]  | 1.690020920  | 4.10059174  | -0.8286511 | -2.7204201 | -0.89839668 | 0.477869956  |
| [9,]  | -2.017591018 | -0.19055562 | 1.2282190  | -0.1591693 | -1.81724286 | -0.504760633 |
| [10,] | 2.684027310  | -0.28238921 | -2.7053968 | -0.6373616 | 1.24913045  | 0.114945320  |
| [11,] | 1.602080176  | 1.73848677  | -0.7560946 | -0.1669273 | 1.46214681  | -1.241067456 |
| [12,] | 3.864282071  | 0.91902758  | -3.0506154 | -0.3116270 | 0.14774890  | -1.707816527 |
| [13,] | -1.537400621 | 0.79553155  | 0.2734490  | 2.2991348  | -1.33509038 | 1.020459263  |
| [14,] | -3.180580364 | -1.55370059 | -1.1043790 | 0.3994455  | -0.21820422 | -0.416321249 |
| [15,] | 0.251424671  | -0.87357501 | 0.8416185  | 1.3248859  | 1.62509470  | -1.332946815 |
| [16,] | 2.892895033  | -0.54107588 | -0.4987349 | 0.5314171  | -0.40738282 | -0.008629769 |
| [17,] | 2.025139659  | 0.87994104  | 2.1794942  | 1.4021183  | -1.10315491 | -1.185333328 |
| [18,] | -4.509205462 | 2.93089784  | -2.9570995 | 3.4152682  | 0.56608967  | 1.421754123  |
| [19,] | -0.363910804 | 0.59536235  | 1.0533136  | -0.1849003 | -0.12823425 | -0.808435775 |
| [20,] | 0.513833400  | 1.34820539  | 1.7227901  | -1.3132714 | -1.04480943 | 0.987729132  |

|       |              |             |             |               |              |              |
|-------|--------------|-------------|-------------|---------------|--------------|--------------|
| [21,] | 0.671318696  | -1.57314535 | 0.3233421   | -0.3775963    | 1.07805278   | 0.122220926  |
| [22,] | -1.199789658 | -0.75223096 | 0.2125650   | -1.9753352    | -0.38988416  | 2.577574944  |
| [23,] | 2.699925571  | -4.05301101 | -2.2580650  | 0.1089181     | -2.08232998  | 0.035139868  |
| [24,] | 3.671607537  | -2.76308860 | 2.3108314   | 1.9763565     | -0.12222446  | 1.145347105  |
| [25,] | 1.708338444  | 1.14870366  | 0.1941423   | -0.8978228    | 0.53891069   | 1.218476886  |
| [26,] | 0.697044049  | -0.09475043 | -0.9235260  | 0.5556583     | 1.51777743   | -0.726583470 |
| [27,] | -3.637428809 | -1.19350763 | -0.2178772  | -1.3363292    | -0.26449368  | -3.193959083 |
| [28,] | 0.919217152  | 1.59671229  | 1.9213986   | 0.1968770     | 0.58065254   | 0.264448784  |
| [29,] | -0.012679393 | 2.32518659  | 0.2155462   | 0.3501560     | -0.47396645  | 0.658332807  |
| [30,] | -2.075112687 | -0.04538414 | 0.2156446   | -1.0632633    | 0.42197992   | 0.038464825  |
| [31,] | -2.320188992 | -0.57228262 | -0.1349819  | 2.0161446     | -0.36632609  | -0.152398000 |
| [32,] | 0.005054503  | -1.40748778 | -2.5517407  | -0.5548639    | -3.02867573  | 0.937512584  |
| [33,] | 0.061169986  | -0.15050467 | 1.6836625   | -0.7372732    | 0.53018221   | 0.408905305  |
|       |              |             |             |               |              |              |
| [1,]  | PC7          | PC8         | PC9         | PC10          | PC11         | PC12         |
| [2,]  | 0.94163613   | 1.01611660  | 0.65086100  | 0.7049538165  | -0.12679861  | 1.07464589   |
| [3,]  | 2.49608290   | -1.03057653 | 0.99789650  | -0.6955702786 | 0.62054036   | 0.43728658   |
| [4,]  | -1.14833452  | -0.13690605 | 1.56172193  | -1.0103974039 | -0.22663448  | -0.62688596  |
| [5,]  | 1.55864777   | -2.37965647 | -1.88594310 | 0.1842266066  | 0.56734997   | -0.37865892  |
| [6,]  | -0.90420171  | -1.46124279 | -0.41077774 | 0.7877696717  | 0.87274235   | -0.40364293  |
| [7,]  | -0.48508037  | -0.39417790 | -1.03586330 | -0.5943650505 | -0.20101933  | 0.39114013   |
| [8,]  | 0.92263413   | 0.27756052  | 0.64943390  | 0.6951735358  | 0.85851959   | -0.18227075  |
| [9,]  | 1.01619589   | 1.42931518  | -0.35783368 | -0.0705759673 | 0.38608546   | -0.45604663  |
| [10,] | -3.06056251  | 0.01542140  | 0.97910235  | -0.3689951357 | -0.81042854  | 0.29905809   |
| [11,] | -1.15019858  | -2.22756355 | 1.92108719  | 1.1858684843  | 0.03051034   | -0.44295070  |
| [12,] | -1.38076963  | 0.10114808  | -1.28404266 | 0.4344189806  | 0.24591922   | -0.18193776  |
| [13,] | 0.63745807   | -0.33245072 | -0.46272040 | -0.4907654143 | -0.49725530  | 0.06959804   |
| [14,] | 0.50021492   | 1.64298989  | -0.02204408 | -0.0008218307 | -0.32919047  | -1.47757160  |
| [15,] | 0.96910614   | -0.33466161 | 1.43687692  | -0.8126002209 | -1.05968338  | -0.42468108  |
| [16,] | 0.80890447   | -0.52662610 | -0.88626822 | 0.4773237243  | -0.83842767  | 0.73180898   |
| [17,] | 0.99711436   | 1.01401737  | -0.40202982 | 0.1817443327  | -0.53794142  | -0.60055239  |
| [18,] | 0.20967322   | -0.58311856 | 0.70515256  | -0.0396632716 | 0.45009864   | 0.68839301   |
| [19,] | -0.81537520  | -0.85666393 | -0.94273181 | 0.0414522567  | 0.07186325   | 0.63521158   |
| [20,] | -0.39788697  | 0.90638787  | -0.41749861 | 0.8035047424  | -0.32339287  | -0.65225431  |
| [21,] | -1.71530837  | -0.05810388 | -0.44792612 | 0.6080155185  | 0.04950397   | 0.22979058   |
| [22,] | -0.17473258  | -0.49430919 | -0.74983375 | -2.3172182635 | -0.86711224  | -0.89907281  |
| [23,] | 0.96335825   | 0.59588805  | -0.69173585 | 0.7875438802  | -1.58060139  | 0.69952356   |
| [24,] | -0.79226369  | 0.31761740  | -1.80001880 | -0.5911334313 | 0.33709088   | 0.50027519   |
| [25,] | 0.66242150   | 0.79456398  | 1.28229789  | 1.1369072252  | 0.36494912   | 0.29993320   |
| [26,] | 0.11491096   | -0.67752131 | 1.55591224  | -0.9531517638 | -0.43306504  | 0.96727700   |
| [27,] | -0.97171844  | 2.00236186  | 0.03915553  | 0.0979107090  | 1.01294064   | -0.86467646  |
| [28,] | -0.10713776  | 1.13254478  | 0.06800519  | 1.3104870153  | -0.78844151  | 0.38194366   |
| [29,] | -0.11368363  | -1.13303668 | -0.44515516 | -0.0309968966 | -0.05286304  | -0.17959227  |
| [30,] | 0.03995439   | -0.06103477 | -0.17348145 | 0.3417751499  | -1.22149166  | -0.09463653  |
| [31,] | -0.61338026  | 1.72693492  | -0.07276041 | -1.2660686396 | 1.38200547   | 1.95263797   |
| [32,] | 0.79663725   | 0.35281020  | 0.08109498  | -0.5131215937 | 0.67712222   | -0.13873520  |
| [33,] | -0.05972762  | -1.30023954 | 0.11504776  | 0.5298008516  | 1.06058730   | -0.29503060  |
| [34,] | 0.25541147   | 0.66221148  | 0.44501905  | -0.5534313394 | 0.90651818   | -1.05932657  |
|       |              |             |             |               |              |              |
| [1,]  | PC13         | PC14        | PC15        | PC16          | PC17         |              |
| [2,]  | 1.04550104   | 0.30220825  | -0.26493487 | 0.990718773   | -0.161522621 |              |
| [3,]  | -0.51140324  | 0.23962723  | 0.04514841  | -0.864576510  | 0.021378514  |              |
| [4,]  | -0.45367844  | 0.47042709  | 0.44536737  | 0.117387548   | 1.132044168  |              |
| [5,]  | 0.76623701   | 0.33414241  | -0.06674780 | 0.005890288   | -0.317090528 |              |
| [6,]  | -0.22473438  | 1.02668595  | -1.11036905 | 0.253771120   | 0.573823233  |              |
| [7,]  | -0.21758119  | 0.35417045  | 0.55539524  | 0.380784887   | 0.577014348  |              |
| [8,]  | -1.13189958  | -0.40654870 | 0.30620976  | 0.183130027   | -0.149934970 |              |
| [9,]  | 0.48772693   | -0.20754494 | 0.30040817  | 0.432882057   | -0.190155729 |              |
| [10,] | 0.47865039   | 0.06044313  | 0.29758301  | 0.065987000   | -0.412015158 |              |
| [11,] | -0.03562393  | 0.19804842  | 0.33881205  | -0.396898429  | -0.531313384 |              |
| [12,] | -0.90180692  | -0.34546312 | -0.75620771 | 0.625598888   | -0.544476288 |              |
| [13,] | 0.30509441   | 0.08813338  | -0.58059025 | -0.035026694  | 0.625823661  |              |
| [14,] | -0.73594105  | 0.53970030  | -0.79585891 | 0.555871566   | -0.252855797 |              |
| [15,] | -0.21555707  | -0.91839176 | -1.10063830 | -0.346724418  | -0.207470713 |              |
| [16,] | 0.72072562   | -0.64024769 | -0.07151878 | -0.333963902  | 0.392495606  |              |
| [17,] | 0.13959352   | 0.68049473  | 0.84486635  | -0.265106061  | -0.372669045 |              |
| [18,] | -1.02024905  | -0.87052303 | 0.58413566  | 0.839988817   | 0.250265142  |              |

```
[18,] 0.77126745 -0.26031481 0.33705965 -0.046675725 0.049282576
[19,] 0.70815779 0.48976813 0.01791635 -0.847195425 -0.125274155
[20,] 0.62144737 -0.75001136 -0.14219657 -0.508865779 0.007346965
[21,] 0.41210426 -0.10629725 0.27929458 0.308950842 -0.697452881
[22,] -0.59841689 0.14146264 0.02533706 -0.177525294 0.552971887
[23,] -0.77530358 -0.21170311 -0.19235313 -0.271578336 -0.009956336
[24,] 0.33558108 -0.22233178 -0.25908437 -0.300951777 -0.380190914
[25,] 0.64085526 0.78994926 -0.57225157 0.361706690 -0.160414591
[26,] 0.13507319 0.08748313 0.02580035 -0.785383761 0.456930349
[27,] -0.10431199 0.13635919 0.47578097 0.572230799 -0.118992564
[28,] -1.18521749 -0.05682740 0.35960572 -0.270080750 -0.621317543
[29,] -0.75999067 -0.33173999 0.05271739 -0.414813487 0.312481307
[30,] -0.34096948 0.30426194 -0.35124709 -0.539719364 -0.359452130
[31,] 0.15560337 0.44934089 0.86956307 0.051061891 -0.041440723
[32,] 0.33869730 -0.18428438 0.18487185 0.420292080 0.248554770
[33,] 1.13036896 -1.18047718 -0.08187461 0.238832439 0.453583546

      PC18      PC19      PC20      PC21
[1,] 0.512534055 0.403637255 -0.2288274019 -0.180597545
[2,] 0.929440714 0.048122004 0.2051146098 -0.271152883
[3,] 0.403516656 -0.074179505 -0.2401958070 -0.046316428
[4,] -0.380413260 -0.016801083 0.0236080284 0.075154120
[5,] -0.121436088 0.216713273 -0.2532722027 0.163180806
[6,] 0.016193332 -0.176022291 0.8096079044 -0.052570563
[7,] -0.041349452 0.491356049 0.1502223084 -0.205615503
[8,] -0.169826040 -0.182268894 -0.0271317045 0.331935144
[9,] -0.170416032 0.311027931 0.4390800414 -0.032482679
[10,] -0.228929235 -0.629795038 -0.1373141246 0.072909759
[11,] 0.223418908 -0.399313078 0.1457129024 -0.346000339
[12,] -0.361282411 0.136577651 0.1213875759 0.000801914
[13,] 0.048394699 -0.297878346 0.0857017322 -0.076965204
[14,] -0.556266739 0.183329956 0.1575056075 0.005626377
[15,] 0.003248088 -0.275646771 -0.0488034493 -0.010278083
[16,] -0.569143575 0.610124017 0.0001699299 -0.165370989
[17,] -0.876273716 0.145493206 -0.3407035497 -0.133229363
[18,] 0.139800530 0.266416571 -0.0612808299 0.138282533
[19,] -0.288699113 -0.005113258 0.1267147465 -0.263464768
[20,] 0.261595957 0.070258777 -0.3355360658 -0.432480871
[21,] 0.478432361 -0.146679334 -0.3985086902 -0.071646635
[22,] -0.090751620 -0.213675088 -0.1815364089 -0.142428632
[23,] 0.266768773 0.195774478 -0.1770461524 0.191720088
[24,] 0.354865033 -0.239914062 0.0937794098 0.244839284
[25,] -0.161337294 0.208473109 -0.0508044281 0.105092550
[26,] -0.040945263 0.377943641 -0.1672829144 0.091569499
[27,] 0.419873948 -0.083337959 0.0482601803 0.312179849
[28,] 0.322418197 0.609269862 -0.0166547989 0.337744117
[29,] 0.091695107 -0.301130070 -0.0161094691 0.322694249
[30,] -0.412731002 -0.414631089 0.0294813311 0.084790553
[31,] -0.298987891 -0.721677547 -0.2130771535 -0.111634161
[32,] 0.187128445 -0.059177362 0.2716590271 -0.102730154
[33,] 0.109463929 -0.037277006 0.1860798156 0.166443959
```

So to combine them with the original variables, do this:

```
CODE
lecturerpcscores ← cbind(lecturers, pca1$x)
```

## Biplot

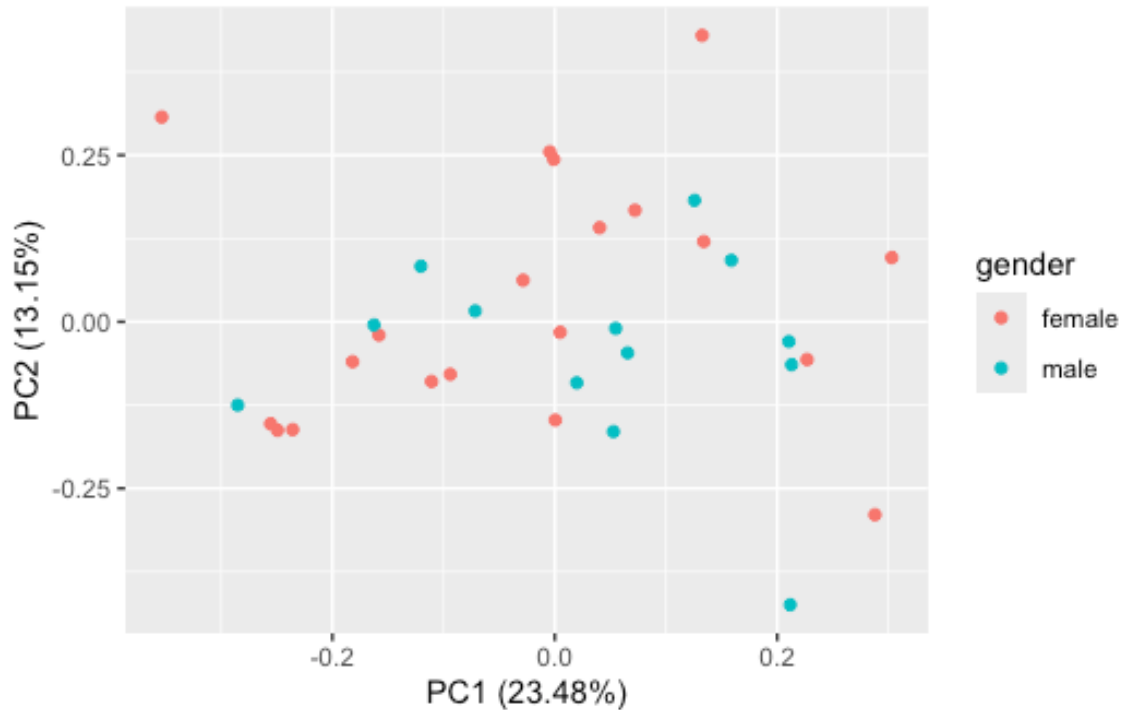
We can produce a biplot in ggplot2, using the factor gender as the colour:

CODE

```
autoplot(pca1, data = lecturerpcscores, colour = 'gender')
```

OUTPUT

```
Warning: `aes_string()` was deprecated in ggplot2 3.0.0.  
i Please use tidy evaluation idioms with `aes()``.  
i See also `vignette("ggplot2-in-packages")` for more information.  
i The deprecated feature was likely used in the ggfortify package.  
Please report the issue at <https://github.com/sinhrks/ggfortify/issues>.
```



We could add the loadings, but they are a bit messy, best to look at loadings or rotations tables.

You can make the plot using different symbols, change background etc through reading the ggplot2 documentation.

## ANOVA on PC scores

You can use this file to do the ANOVAs, you will have to do yourself. For example for differences in PC scores for gender:

CODE

```
A0V1 <- aov(lecturerpcscores$PC1 ~ gender, data = lecturerpcscores)  
summary(A0V1)
```

OUTPUT

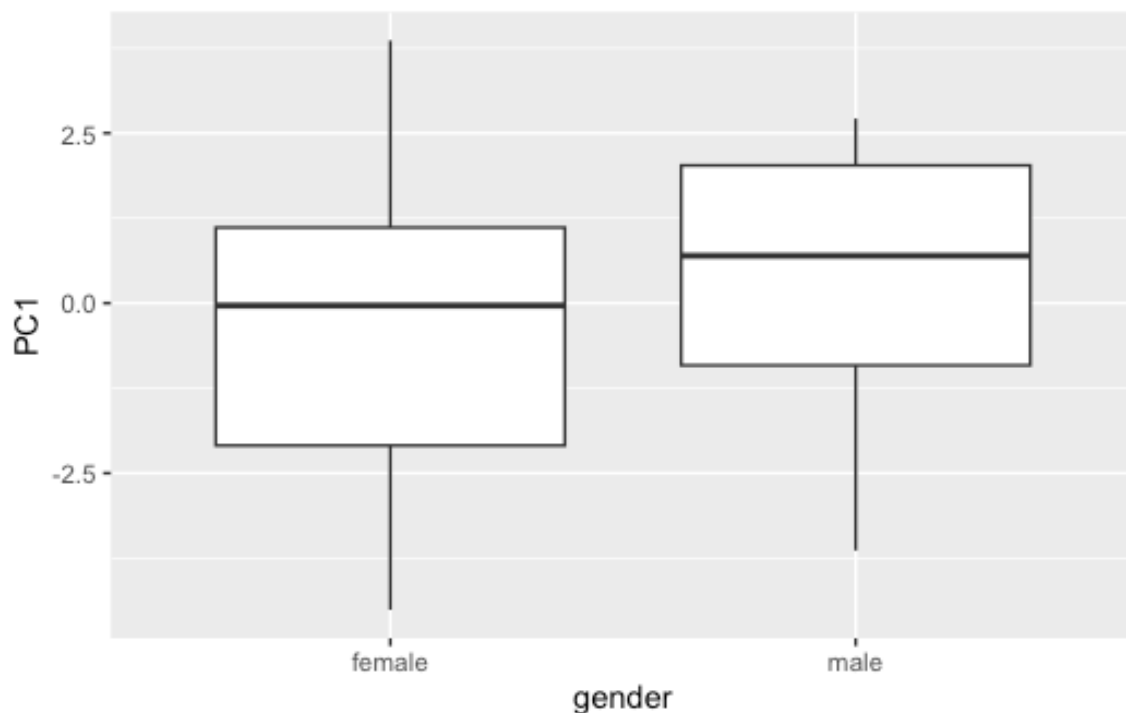
|           | Df | Sum Sq | Mean Sq | F value | Pr(>F) |
|-----------|----|--------|---------|---------|--------|
| gender    | 1  | 4.6    | 4.596   | 0.93    | 0.342  |
| Residuals | 31 | 153.2  | 4.942   |         |        |

You can do something similar for your projects and regression, using your PC scores as the predictors.

## Boxplot

To do a boxplot of variables use ggplot2:

```
CODE
ggplot(lecturerpcscores, aes(x = gender, y = PC1)) +
  geom_boxplot()
```



## Factor Analysis with Varimax Rotation

To do a Varimax rotation on your principal components, follow these commands. Do a factor analysis with the rotation being Varimax:

```
CODE
fa1 <- factanal(lecturers[, 8:28], 10, rotation = "varimax")
fa1
```

# OUTPUT

Call:

```
factanal(x = lecturers[, 8:28], factors = 10, rotation = "varimax")
```

Uniquenesses:

|                         |                   |                         |
|-------------------------|-------------------|-------------------------|
| accessible              | attractive        | wide_range              |
| 0.381                   | 0.005             | 0.005                   |
| teaching.based_research | Dress_nicely      | focus_on_complex_issues |
| 0.005                   | 0.283             | 0.332                   |
| gives_example           | gives_out_note    | sense_of_humour         |
| 0.005                   | 0.448             | 0.473                   |
| mark_easily             | moves_around      | offers_lecture_notes    |
| 0.005                   | 0.294             | 0.141                   |
| clear_objective         | solicit_questions | text_book               |
| 0.244                   | 0.005             | 0.310                   |
| primary_research        | own_research      | teach_in_team           |
| 0.310                   | 0.237             | 0.548                   |
| understand_principles   | use_a_pointer     | well_respected          |
| 0.584                   | 0.067             | 0.005                   |

Loadings:

|                         | Factor1 | Factor2 | Factor3 | Factor4 | Factor5 | Factor6 | Factor7 |
|-------------------------|---------|---------|---------|---------|---------|---------|---------|
| accessible              |         | -0.583  | -0.426  | 0.222   | -0.110  |         | -0.130  |
| attractive              |         | 0.961   | 0.106   |         |         |         | 0.158   |
| wide_range              | -0.177  |         |         | 0.952   |         |         |         |
| teaching.based_research |         |         | -0.356  | 0.374   | -0.114  |         |         |
| Dress_nicely            |         | 0.829   | 0.100   |         |         |         |         |
| focus_on_complex_issues | 0.116   |         |         | 0.674   |         | -0.163  | -0.141  |
| gives_example           |         | 0.122   | 0.112   |         | 0.955   | 0.163   | 0.132   |
| gives_out_note          | 0.362   | 0.290   | 0.216   | -0.196  | 0.305   | 0.125   | 0.261   |
| sense_of_humour         | 0.127   | 0.229   | 0.599   | 0.125   |         |         | 0.108   |
| mark_easily             | 0.150   | 0.298   | 0.143   |         |         |         | 0.910   |
| moves_around            | 0.326   | 0.244   | 0.186   | -0.102  | -0.155  |         | -0.194  |
| offers_lecture_notes    | 0.362   | 0.240   | 0.333   |         | 0.667   |         | -0.232  |
| clear_objective         | 0.768   | 0.109   | 0.343   |         | 0.169   |         |         |
| solicit_questions       | 0.880   | -0.139  | 0.130   | 0.109   | -0.156  |         | 0.169   |
| text_book               | 0.611   |         | -0.141  | -0.152  | 0.166   | 0.202   | 0.427   |
| primary_research        |         |         |         |         |         |         |         |
| own_research            | 0.100   |         |         | 0.129   |         | -0.112  | -0.102  |
| teach_in_team           | 0.154   |         | -0.103  |         | 0.155   | 0.614   |         |
| understand_principles   | 0.608   |         |         |         |         | 0.155   |         |
| use_a_pointer           | 0.164   | 0.146   | 0.902   | -0.142  | 0.165   |         |         |
| well_respected          | 0.133   |         | 0.287   |         |         | 0.931   | 0.101   |

|                         | Factor8 | Factor9 | Factor10 |
|-------------------------|---------|---------|----------|
| accessible              |         |         |          |
| attractive              |         | -0.121  |          |
| wide_range              | 0.102   | 0.137   | -0.126   |
| teaching.based_research | 0.158   | 0.819   |          |
| Dress_nicely            |         |         |          |
| focus_on_complex_issues |         |         | 0.369    |
| gives_example           |         |         |          |
| gives_out_note          | 0.244   |         | -0.120   |
| sense_of_humour         |         | -0.226  |          |
| mark_easily             | -0.113  | -0.106  |          |
| moves_around            | 0.429   | 0.444   | 0.217    |
| offers_lecture_notes    | -0.159  |         | 0.167    |
| clear_objective         |         |         |          |
| solicit_questions       | 0.326   |         |          |
| text_book               | -0.146  |         |          |
| primary_research        | 0.213   |         | 0.790    |
| own_research            | 0.784   | 0.119   | 0.255    |
| teach_in_team           |         |         |          |
| understand_principles   |         | 0.110   |          |
| use_a_pointer           | 0.109   |         |          |



```

well_respected      -0.100

      Factor1 Factor2 Factor3 Factor4 Factor5 Factor6 Factor7 Factor8
SS loadings      2.654  2.398  1.996  1.705  1.667  1.423  1.325  1.137
Proportion Var   0.126  0.114  0.095  0.081  0.079  0.068  0.063  0.054
Cumulative Var   0.126  0.241  0.336  0.417  0.496  0.564  0.627  0.681
      Factor9 Factor10
SS loadings      1.039  0.970
Proportion Var    0.049  0.046
Cumulative Var    0.731  0.777

Test of the hypothesis that 10 factors are sufficient.
The chi square statistic is 28.56 on 45 degrees of freedom.
The p-value is 0.973

```